

COSH: Closed-Form Onset Search via Hyperbolic Response

Methodology for pre-flight critical-moment identification of a multirotor from ground-contact tip-over excitation

This document specifies the complete identification methodology, from raw signals to the take-off feed-forward moment. Every stage mirrors the implementation in `critical_value_getter_pieewise.py`; the companion analyses (model-fidelity bounds, excitation design, ground-effect bracket) are documented in `fidelity_section.tex` and are only referenced here.

1 Problem setting and notation

A multirotor rests on its landing gear. Ramping the commanded roll (pitch) moment tilts the vehicle until it begins to rotate about the landing-gear contact line — the *pivot*. The applied moment at that *rotation onset* is the critical moment M_{crit} ; comparing the two tip directions yields the CoM offset and the feed-forward moment M_{ff} that compensates take-off. No rig, no motion capture and no stable in-flight data are used.

Symbol	Meaning	Value	Provenance
W, z_{CoM}	weight, CoM height	31.59 N, ~ 0.3 m	measured / conservative
$l_{p,\phi}, l_{p,\theta}$	pivot offsets (roll, pitch)	0.140, 0.110 m	conservative geometry
J_P	inertia about the pivot line	—	never used numerically
C_T	thrust map, $T_i = C_T \Omega_i^2$	1.3175×10^{-7}	bench calibration
M_{thr}	window threshold	0.01 N m	noise floor
M_{cap}	allocator caps (x, y)	2.37, 2.74 N m	allocation feasibility
ε_{max}	ramp-slope gate	3%	sensitivity chain (Sec. 4)
N_{min}	window-sample gate	38	30 pre + 8 post
$\varepsilon_{\text{lin,max}}$	linearity-RMSE gate	30 mN m	10 $\times$ noise floor
$N_{\text{pre}}, N_{\text{post}}$	sweep margins	30, 8	baseline / fit minima

Table 1: Design constants. (C_2, K) are *calibrated*, not set (Sec. 6).

2 Contact-phase dynamics and the closed form

About the active pivot line, with the small-angle attitude dependence retained ($\cos \varphi \approx 1$, $\sin \varphi \approx \varphi$), the roll/pitch dynamics read

$$J_P \dot{\omega} = M(t) + f l_p - W (l_p + \lambda_{\text{off}}) + W z_{\text{CoM}} \varphi, \quad (1)$$

where M is the commanded body moment, f the collective thrust and λ_{off} the CoM offset along the excited axis. Below the critical value the vehicle is static. At the onset the net moment vanishes, which supplies *two* anchoring conditions,

$$\omega(t_{\text{crit}}) = 0, \quad \dot{\omega}(t_{\text{crit}}) = 0. \quad (2)$$

Past the onset gravity feeds back positively: the linearized dynamics have *real* eigenvalues $\pm C_2$ and, with the measured moment ramp $M(t) = M_{\text{crit}} + \dot{M} \tau$ in onset-relative time $\tau = t - t_{\text{crit}}$, the exact solution of (1)–(2) collapses to the hyperbolic closed form

$$\omega(\tau) = C_1 (\cosh(C_2 \tau) - 1), \quad C_1 = K \dot{M}, \quad K = \frac{1}{W z_{\text{CoM}}}, \quad C_2 = \sqrt{\frac{W z_{\text{CoM}}}{J_P}} \quad (3)$$

for $\tau \geq 0$ (and $\omega = 0$ for $\tau < 0$). Because $J_P C_2^2 = W z_{\text{CoM}}$, the amplitude C_1 is *fixed* by the measured ramp rate: with (C_2, K) pinned as rig constants and the baseline fixed by continuity, **no shape parameter is free per run**. The time-quadratic model of the original submission is the $C_2 \rightarrow 0$ limit of (3) — it discards the gravity term that creates the instability, with 14–75% truncation error over the windows actually used — and is retained only as a baseline.

3 Signals and excitation window

Per run, the moment is reconstructed from the measured rotor speeds, $M(t) = C_T \sum_i b_{a,i} \Omega_i^2(t)$, and $\omega(t)$ is taken from the odometry rate. The window end is $i_{\text{end}} = \arg \max_t |M|$, *truncated at the first sample exceeding the allocator cap* M_{cap} : past it the thrust allocation saturates, the executed ramp is no longer linear, and the cosh family does not apply. (On the reference dataset the cap never binds; peak $|M| \leq 1.7$ N m.) The window start is the *last* sub-threshold sample before the peak, walking backwards — robust to spin-up blips that cross the threshold long before the actual ramp.

4 Onset-free run gates

Before any onset is estimated, each run must pass three gates evaluated on the measured moment trace alone (hence free of circularity):

1. **Ramp-slope error** $|\varepsilon| \leq 3\%$, with $\varepsilon = (|\dot{M}| - \dot{M}_{\text{cmd}})/\dot{M}_{\text{cmd}}$ and $\dot{M}$ the least-squares slope over the window. A 3% error in $\dot{M}$ perturbs $C_1 = K\dot{M}$ by 3% — about $7\times$ below the $\pm 20\%$ level the sensitivity analysis shows harmless.
2. **Window samples** $N_{\text{full}} \geq 38$: a necessary condition for any onset position (30 pre-onset for the baseline and sweep, 8 post-onset for the fit).
3. **Linearity RMSE** ≤ 30 mN m about the run’s own best-fit line: a fault detector for aborted/stepped ramps, set at $10\times$ the execution noise floor and above the healthy-run maximum.

The gates run *before* calibration so a poorly executed ramp cannot contaminate the dataset-coupled constraint $C_1 = K\dot{M}$.

5 Onset detection

With (C_2, K) pinned, detection reduces to an exhaustive sweep of the single remaining unknown — the onset index. Each candidate is a plain arithmetic evaluation: no optimizer, no initial guess, no seed model.

Algorithm 1 COSH onset detection (single run)

Require: windowed samples $\{(t_i, \omega_i)\}_{i=1}^N$; moment $M(t)$; calibrated (C_2, K) ; measured ramp slope $\dot{M}$

Ensure: onset index j^* ; critical moment M_{crit}

- 1: $C_1 \leftarrow K\dot{M}$ ▷ onset conditions (2) fix the amplitude
 - 2: **for** $j = N_{\text{pre}}, \dots, N - N_{\text{post}}$ **do** ▷ exhaustive sweep of the single unknown
 - 3: $C \leftarrow \text{median}\{\omega_i : i < j\}$ ▷ baseline by continuity
 - 4: $\text{cost}(j) \leftarrow \sum_{i < j} (\omega_i - C)^2 + \sum_{i \geq j} (\omega_i - [C_1(\cosh C_2(t_i - t_j) - 1) + C])^2$
 - 5: **end for**
 - 6: $j^* \leftarrow \arg \min_j \text{cost}(j)$; $M_{\text{crit}} \leftarrow M(t_{j^*})$
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Remark 1 *Because the model curve is fully determined before the post-onset data are seen, goodness of fit is a prediction test, not flexibility: across all gate-passing runs the normalized post-onset residual is $\approx 6\%$ (gyro-noise dominated) and invariant over the twelvefold ramp-rate range — the empirical certificate that the response stays in the cosh family.*

6 Rig-constant calibration

Algorithm 1 presupposes (C_2, K) . Although both have geometric expressions (3), they are *not* computed from geometry: z_{CoM} and especially J_P are not precisely known a priori, and — more importantly — the constants that make (3) reproduce the measured response are *effective* values that absorb the state-linear contributions of the unmodeled effects (the local slope of the gravity nonlinearity and the ground-effect spring). Both are calibrated once per dataset.

Joint residual minimization is not a usable criterion: in the cost of Algorithm 1, an earlier onset with a smaller amplitude trades off against a later onset with a larger one, so the summed residual is nearly degenerate along a ridge in the (C_2, K) plane and its minimizer is not physically meaningful. We calibrate instead by *physical self-consistency*: a static threshold must not depend on how fast it was approached. With runs at ramp rates $r \in \{0.10, \dots, 1.20\}$ N m/s,

$$\text{score}(C_2, K) = \sum_{\text{dir} \in \{+, -\}} \text{CV}(\{M_{\text{crit}}^{(r)}(C_2, K)\}_r), \quad \text{CV} = \text{std}/|\text{mean}|, \quad (4)$$

where $M_{\text{crit}}^{(r)}$ is the output of Algorithm 1 on the run with commanded rate r .

Search ranges. The admissible box is set so that it covers every physically plausible constant with margin. For $K = 1/(W z_{\text{CoM}})$, the coarse range $K \in [0.10, 0.50]$ (N m)^{−1} corresponds to $z_{\text{CoM}} \in [0.06, 0.32]$ m at the vehicle weight — bracketing the plausible CoM heights — with the refinement allowed to move within the wider clip box $[0.05, 0.70]$ (z_{CoM} up to 0.63 m). For $C_2 \in [3, 8]$ rad/s, the implied pivot inertia $J_P = 1/(K C_2^2)$ spans roughly $[0.03, 1.1]$ kg m², bracketing the parallel-axis estimate $J + m z_{\text{CoM}}^2 \approx 0.2$ – 0.35 kg m² by a factor of several on both sides.

Algorithm 2 Rig-constant calibration by ramp invariance (coarse-to-fine)

Require: gated runs at rates $r \in \{0.10, \dots, 1.20\}$ N m/s, both directions; box $C_2 \in [3, 8]$ rad/s, $K \in [0.05, 0.70]$ (N m)^{−1}

Ensure: calibrated pair (C_2^*, K^*)

- 1: **function** SCORE(C_2, K)
 - 2: **for** each gated run r **do**
 - 3: $M_{\text{crit}}^{(r)} \leftarrow$ Algorithm 1 with (C_2, K) pinned ▷ stride-3 sweep
 - 4: **end for**
 - 5: **return** $\sum_{\text{dir} \in \{+, -\}} \text{CV}(\{M_{\text{crit}}^{(r)}\}_r)$ ▷ ramp invariance, Eq. (4)
 - 6: **end function**
 - 7: **coarse:** evaluate SCORE($\cdot$) on the 11×11 grid $C_2 = 3.0:0.5:8.0$, $K = 0.10:0.04:0.50$
 - 8: $(C_2^b, K^b) \leftarrow$ arg min over the coarse grid
 - 9: **fine:** evaluate SCORE($\cdot$) on a 9×9 uniform grid over $[C_2^b \pm 0.5] \times [K^b \pm 0.04]$, clipped to the box ▷ resolution 0.125 rad/s, 0.01 (N m)^{−1}
 - 10: $(C_2^*, K^*) \leftarrow$ arg min over the fine grid
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Coarse-to-fine procedure. Algorithm 2 summarizes the search. The coarse scan evaluates (4) on an 11×11 grid ($\Delta C_2 = 0.5$ rad/s, $\Delta K = 0.04$ (N m)^{−1}; 121 evaluations). A single refinement pass then re-scans a 9×9 uniform grid spanning $\pm$ one coarse step about the coarse minimizer (± 0.5 rad/s, ± 0.04 (N m)^{−1}, clipped to the box), i.e. final resolutions of 0.125 rad/s and 0.01 (N m)^{−1} — so the reported constants are not quantized to the coarse step. One pass suffices because the objective is flat along the (C_2, K) ridge; it is also *piecewise constant* (the onset is an integer index, so small parameter changes leave M_{crit} unchanged), which is why gradient-based or simplex optimizers stall on it, and a population-based global search (differential evolution) was found to cost about $4 \times$ more for no measurable gain. During calibration the onset sweep of Algorithm 1 is subsampled by a stride of 3 samples for speed; the final identification runs at full resolution with the selected pair. Each evaluation is an arithmetic sweep, so no iterative optimizer is involved anywhere in the pipeline.

Remark 2 The objective (4) retains a shallow valley, so the individual constants are only loosely determined: $W z_{\text{CoM}} = 1/K$ and $J_P = 1/(K C_2^2)$ are order-of-magnitude sanity checks, not precision measurements. What the criterion pins down — by construction — is the identified onset: M_{crit} is robust along the valley, and a $\pm 20\%$ mis-specification of either constant moves the identified CoM by less than 0.9 mm.

7 Aggregation and deliverables

Directional means over the gated runs give $\bar{M}_+$ and $\bar{M}_-$. The take-off feed-forward moment is the *pivot-free average*

$$M_{\text{ff}} = \frac{1}{2} (\bar{M}_+ + \bar{M}_-), \quad (5)$$

in which the pivot-arm terms $\pm(W - f) l_p$ — and any thrust-proportional contamination acting through the arm l_p , ground effect included — cancel by antisymmetry, while landing-gear asymmetry is deliberately *retained*: the same contact-phase moment acts at lift-off, where M_{ff} compensates it. The CoM offset follows from the moment-to-offset conversion of M_{ff} and is used for validation against load-cell ground truth only; M_{ff} itself is the quantity applied in flight.

8 Reproducibility

Every number in this document is reproducible from the repository: `critical_value.getter.piecewise.py` (pipeline, gates, caps, calibration, Algorithm 1); `analysis/ramp_quality.py` (gates); `analysis/sensitivity.py` ($\pm 20\%$ robustness); `analysis/cosh_fidelity.py` (prediction-test residuals); `analysis/excitation_angle_design.py` (rate/cutoff design); `analysis/ge_linearity.py`, `ge_trajectory.py` (ground-effect bracket); `analysis/tiltcap_ablation.py` (window ablation).